Noise Robust Radar HRRP Target Recognition Based on Multitask Factor Analysis With Small Training Data Size

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***Abstract*—A factor analysis model based on multitask learning (MTL) is developed to characterize the FFT-magnitude feature of complex high-resolution range profile (HRRP), motivated by the problem of radar automatic target recognition (RATR). The MTL mechanism makes it possible to appropriately share the information among samples from different target-aspects and learn the aspect-dependent parameters collectively, thus offering the potential to improve the overall recognition performance with small training data size. In addition, since the noise level of a test sample is usually different from those of the training samples in the real application, another contribution is that the proposed framework can update the noise level parameter in the FA model to adaptively match that of the received test sample. Efficient inference is performed via variational Bayesian (VB) for the proposed hierarchical Bayesian model, and encouraging results are reported on the measured HRRP dataset with small training data size and under the test condition of low signal-to-noise ratio (SNR).**

***Index Terms*—Factor analysis (FA), hierarchical Bayesian modeling, high-resolution range profile (HRRP), multitask learning (MTL), radar automatic target recognition (RATR), variational Bayes (VB).**

I. Introduction

RADAR high-resolution range profile (HRRP) has received intensive attention from the radar automatic target recognition (RATR) community [5], [7]–[10], [12], [14], [16]–[18], [20], [25], [27]. According to the definition of HRRP given in these literatures, an HRRP is a *real* vector, which is composed of the amplitude of the coherent summations of the complex returns from target scatterers in each range cell. As shown in [7], [8], [10], [12], [14], [17], [18], [20], [27], statistical recognition is an efficient method for real HRRP based RATR. By statistical recognition is meant that the class membership of a measurement HRRP feature vector $\boldsymbol{x}$ is determined based on the posterior probabilities of all class memberships $p(c\,|\,\boldsymbol{x})$ ($c \in \{1, 2, \ldots, C\}$ denotes the class membership). According to Bayes theorem, we have $p(c\,|\,\boldsymbol{x}) \propto p(\boldsymbol{x}\,|\,c)p(c)$, where $p(\boldsymbol{x}\,|\,c)$ is the class-conditional likelihood and $p(c)$ is the prior class probability. Since it is assumed that the targets have equal *a prior* probabilities of occurrence, estimation of the posterior probability is turned into estimation of the class-conditional likelihood, i.e., statistical modeling.

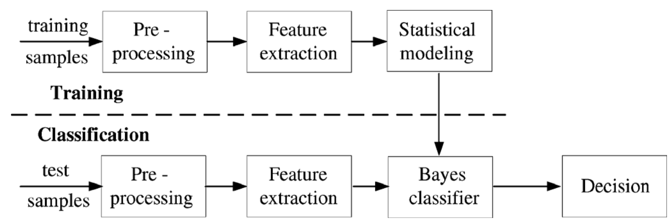

Fig. 1. A typical flow chart of statistical recognition.

A typical statistical recognition scheme is depicted in Fig. 1. As we can see, there are two stages, i.e., training stage and classification stage, in the statistical recognition procedure, both of which include preprocessing and feature extraction steps. For radar HRRP recognition, as discussed in [9] and [10], some preprocessing techniques are needed to deal with the target-aspect, time-shift, and amplitude-scale sensitivity problems.

Feature extraction from HRRP is a key step in the recognition system. Real HRRP sample is a widely used feature in the radar HRRP statistical recognition area [7], [8], [10], [12], [14], [18], [20]. References [16] and [25] investigate the bispectra feature extracted from real HRRP data, and [9] further compares the recognition performance of various high-order spectra features also extracted from real HRRP data. In [17] and [27], some superresolution algorithms are employed to extract the location information of predominant scatterers from real HRRP data as feature. Nevertheless, some information contained in the HRRP samples is inevitably lost during the dimensional reduction procedure. As demonstrated in [11], compared with the real HRRP data, the complex HRRP data, which can usually be directly and easily registered via a high resolution radar system, reserve some phase information useful for target discrimination. Nevertheless, there has been little work done in the field of RATR using complex HRRPs. We believe that the reason is complex HRRPs' strong sensitivity to target position variation, which is referred to as the initial-phase sensitivity. Therefore, the initial-phase invariant feature extraction is a precondition for statistical modeling for complex HRRP data.

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Some parametric statistical models are developed to describe the class-conditional likelihood of real HRRP data, e.g., Gaussian model [14], [18], Gamma model [7], the two-distribution compounded model comprising Gamma distribution and Gaussian mixture distribution [10], factor analysis (FA) model [8], and the local factor analysis (LFA) model [20]. In addition, the hidden Markov model (HMM) has been utilized by [17] and [27] for radar HRRP sequence recognition. Nevertheless, there are two limitations of these statistical models. First, they usually require substantial training data, especially for the complicated statistical models with many unknown parameters, such as FA model in [8] and [20] and HMM model in [17] and [27]. However, due to the sampling rate limitation, it is hard to collect enough training data for each target-aspect of each target. Rather than building models for each data subset associated with different target-aspect individually, it is desirable to appropriately share the information among these related data, thus offering the potential to improve overall recognition performance. If the modeling of one data subset is termed one learning task, the learning of models for all tasks jointly is referred to as multitask learning (MTL) [4]. Our recent work in [12] proposed an HMM-based MTL algorithm (called dynamic MTL-TSB-HMM) for real HRRP samples and the inspiring recognition performance is achieved only with small training data size.[1] Second, all the existing statistical models for HRRP data assume that the training data and test data are obtained under the similar measurement circumstance. In the real application, the training data are usually collected under the condition of high signal-to-noise ratio (SNR) via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the noncooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the severe measurement conditions, such as the large distance between the noncooperative targets and radar. Thus the statistical model for RATR should be modified to match the noise level of the received test sample in the classification stage to obtain the noise-robust recognition performance.

Another kind of noise-robust recognition methods is to extract the noise-robust feature in the feature extraction step. For example, as discussed in [15] and [22], the bispectrum feature of radar signal can suppress additive disturbances that are described by symmetric probability density functions, e.g., additive white Gaussian noise, additive colored Gaussian noise, and additive non-Gaussian noise with the symmetric probability density function. Nevertheless, we focus on noise-robust statistical modeling rather than noise-robust feature extraction in this paper.[2]

[1]In [12] our dynamic MTL-TSB-HMM with 16 HRRP training samples per target-aspect from the measured HRRP dataset can obtain the comparable recognition rates with the traditional statistical models [7], [10], [14], [18] with 128 or more samples per target-aspect in the recognition experiments. In the real application, this means that we can reduce the sampling rate of the radar system, and, therefore, the storage and computation burdens in the training stage can largely decrease.

[2]Although both of [12] and this paper utilize the multitask learning, the two papers focus on solving the different problems in RATR. The dynamic MTL-TSB-HMM in [12] can describe the temporal dependence of multiple HRRPs from a sequence of target-sensor orientations; while the noise-robust MTL-FA in this paper can describe the variation of noise level.

The research reported here seeks a noise-robust statistical recognition method based on MTL for complex HRRP. In this paper, we develop a *noise-robust factor analysis model based on multitask learning* (noise-robust MTL-FA) for the FFT-magnitude feature extracted from complex HRRP data, where the initial-phase and time-shift invariant feature makes it possible to share the same factor loading matrix across different target-aspects of each target, but the latent factor variables are target-aspect dependent. Another contribution is that the proposed framework can update the noise level parameter in the FA model in the classification stage to adaptively match that of the received test sample. In this way, the recognition performance under the test condition of low SNR can be improved. In addition, we can easily reconstruct the test sample with high SNR from the noised one, which can be further utilized to other applications, e.g., analysis on target characteristic, information fusion and dataset expansion. The form of the proposed hierarchical model allows efficient variational Bayesian (VB) inference, of interest for large-scale problems. The RATR experiments based on measured HRRP dataset show that the proposed model not only has good recognition performance with small training data size but also is more robust to noise than other existing statistical models.

The remainder of the paper is organized as follows. We introduce the MTL-FA model for complex HRRP data in Section II. The model learning and noise-robust classification methods are discussed in Section III. The detailed experimental results based on measured HRRP data are provided in Section IV, followed by conclusions in Section V.

II. Multitask Factor Analysis Model

A. Feature Extraction From Complex HRRP Data

Suppose there are radial displacement and little uniform rotation for a target. If the transmitted signal is $s(t)e^{j\omega_c t}$ ($s(\cdot)$ is the complex envelope and ω_c is the carrier angular frequency) and the returned complex HRRP is denoted as a discrete complex vector, the nth complex returned echo from the lth range cell ($l = 1, 2, \ldots, L$) in baseband is

$$x_l(t,n) = \sum_{i=1}^{V_l} \sigma_{li} s\left(t - \frac{2R_{li}(n)}{c}\right) e^{-j[\frac{4\pi}{\lambda} R_{li}(n) - \theta_{li0}]} \quad (1)$$

where λ denotes the radar wavelength, V_l denotes the number of target scatterers in the lth range cell, σ_{li} and θ_{li0} represent the strength and the initial phase of the ith scatterer in the lth range cell, and $R_{li}(n)$ denotes the radial distance between the radar and the ith scatterer in the lth range cell of the nth returned echo.

If $R(n)$ denotes the radial distance between the target reference center in the nth returned echo and the radar, and L_y denotes the radial length of the target, usually $R(n) \gg L_y$. Due to the target rotation, there are radial displacements for the scatterers. If $\Delta r_{li}(n)$ represents the radial displacement of the ith scatterer in the lth range cell in the nth returned echo, then $R_{li}(n) \overset{R(n)\gg L_y}{\approx} R(n) + \Delta r_{li}(n)$. Since the complex envelope $s(\cdot)$ in a range cell approximates to be unvaried with the radial displacements for all scatterers in the range cell, (1) can be approximated as

$$\begin{aligned} x_l(t,n) &\approx x_l(n) \\ &= \sum_{i=1}^{V_l} \sigma_{li} \exp\left\{-j\left[\frac{4\pi}{\lambda}(R(n)+\Delta r_{li}(n)) - \theta_{li0}\right]\right\} \\ &= e^{-j\frac{4\pi}{\lambda}R(n)} \sum_{i=1}^{V_l} \sigma_{li} \exp\left\{-j\left[\frac{4\pi}{\lambda}\Delta r_{li}(n) - \theta_{li0}\right]\right\} \\ &= e^{j\theta(n)} \sum_{i=1}^{V_l} \sigma_{li} e^{j\phi_{li}(n)} \end{aligned} \tag{2}$$

where $\theta(n) = -\frac{4\pi}{\lambda}R(n)$ denotes the initial phase of the nth returned echo, which depends on the target distance and the radar wavelength and does not contain the target discriminative information, and $\phi_{li}(n) = \theta_{li0} - \frac{4\pi}{\lambda}\Delta r_{li}(n)$ denotes the remained echo phase of the ith scatterer in the lth range cell of the nth returned echo.

Then the nth complex HRRP is

$$\begin{aligned} \boldsymbol{x}(n) &= [x_1(n), x_2(n), \ldots, x_L(n)]^{\mathrm{T}} \\ &= e^{j\theta(n)}[\tilde{x}_1(n), \tilde{x}_2(n), \ldots, \tilde{x}_L(n)]^{\mathrm{T}} \end{aligned} \tag{3}$$

where $\tilde{x}_l(n) = \sum_{i=1}^{V_l} \sigma_{li} e^{j\phi_{li}(n)}$ with $l = 1, 2, \ldots, L$. Here the initial phase of the nth complex HRRP sample, i.e., $\theta(n) = -\frac{4\pi}{\lambda}R(n)$, is strongly sensitive to the target position variation. For example, for carrier frequency of 5520 MHz (approximately, the wavelength of 5 cm), if the radial displacement of a target is 2.5 cm, the initial phase difference will be 2π. In the real application, the measurement precision of the target distance usually cannot satisfy the requirement of the initial phase compensation, therefore, the measured complex HRRP data have the initial-phase sensitivity.

Moreover, since an HRRP sample is only a part of received radar echo extracted by a range window, in which target signal is included, the position of the target signal in an HRRP sample can vary with the measurement. This is the time-shift sensitivity. Considering these two sensitivity issues of the complex HRRP data, i.e., the initial-phase sensitivity and time-shift sensitivity, we prefer to extracting a feature which is independent of the complex HRRP's initial-phase and time-shift, and then characterizing such feature via a MTL statistical model in this paper.

References [23] and [24] propose to use FFT-magnitude feature for radar HRRP target recognition. The frequency spectrum $\boldsymbol{\mathcal{F}}(n)$ of a complex HRRP $\boldsymbol{x}(n)$ is

$$\begin{aligned} \boldsymbol{\mathcal{F}}(n) &= \mathrm{FFT}(\boldsymbol{x}(n)) \\ &= e^{j\theta(n)}[\mathcal{F}_1(n), \mathcal{F}_2(n), \ldots, \mathcal{F}_L(n)]^{\mathrm{T}} \end{aligned} \tag{4}$$

where the function $\mathrm{FFT}(\cdot)$ denotes the fast Fourier transform (FFT), and $\mathcal{F}_l(n) = \sum_{l'=1}^{L} \tilde{x}_{l'}(n) e^{\frac{-j2\pi l'(l-1)}{L}}$ with $l = 1, 2, \ldots, L$. From (4) we can see the frequency spectrum $\boldsymbol{\mathcal{F}}(n)$ still suffers from the time-shift sensitivity and initial-phase sensitivity. Nevertheless, according the property of FFT, the time-shift of $\boldsymbol{x}(n)$ can only bring a new phase term to its frequency spectrum. Therefore, same as papers [23], [24], here we only use the amplitude of frequency spectrum $\boldsymbol{y}(n) = |\boldsymbol{\mathcal{F}}(n)|$, i.e., FFT-magnitude, as the feature vector for the following statistical modeling. Obviously, the FFT-magnitude feature is independent of the complex HRRP's time-shift and initial-phase, which is referred to as the time-shift and initial-phase invariant property.

As discussed in [23] and [24], there are many advantages of working in the frequency domain. First, it is found that the frequency domain target signatures are distinct for different aircraft types and at different aspects. As a result, target identification in the frequency domain is just as viable as using conventional real HRRPs in the range domain. Second, since FFT is an information preserving procedure, there is no information lost in the transformation process. Third, the FFT-magnitude feature of complex HRRP has the time-shift and initial-phase invariant property. Finally, for the stepped frequency waveform (SFWF) radar system, it is much easier to get the frequency spectrum of moving targets. Usually, to obtain the HRRP sample from the SFWF, the radial motions along the radar line of sight (RLOS), the velocity and acceleration of the target have to be compensated; otherwise, broadening of the HRRP envelope will occur.

Fig. 2 presents the real HRRPs and corresponding FFT-magnitude feature vectors of 3 airplane targets at the same target-aspect. From Fig. 2 we can see that both of the echo amplitude of the frequency spectra and the corresponding range profiles are discriminative for the three targets.

B. Statistical Modeling

1) Review of the STL-FA Model: According to the scattering center model [13], HRRPs from complex targets yield target signatures that are a strong function of the target-sensor orientation [7]–[11], [14], [17], [18], [20], [23], [24], [27], referred to as target-aspect sensitivity in [8]–[11], [20], and [23]. To deal with the target-aspect sensitivity, the multiaspect HRRP dataset from each target is divided into subsets (as in [8]–[10], [23], and [24]), where the HRRP data in each subset are collected from a target-aspect sector roughly without the scatterers' motion through range cells (MTRC). Thus each subset is defined to be an aspect-frame from the target [8]–[10], [23].

Our previous paper [8] developed an FA model for the real HRRP data from each aspect-frame. The generative model of an L-length HRRP sample in the mth frame of the target c (here $m \in \{1, \ldots, M_c\}$ with M_c denoting the number of aspect-frames of the cth target, and $c \in \{1, 2, \ldots, C\}$ with C denoting the number of targets) is given by $\boldsymbol{x}^{(c,m)}(n) = \boldsymbol{A}^{(c,m)}\boldsymbol{w}_n^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\epsilon}_n^{(c,m)}$, where $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^L$ is the mean vector, $\boldsymbol{A}^{(c,m)} \in \mathbb{R}^{L\times J}$ is the factor loading matrix with J denoting the number of factors and $J < L$, the latent factor variable $\boldsymbol{w}_n^{(c,m)} \in \mathbb{R}^K \sim \mathrm{Norm}(0, \boldsymbol{I}_K)$ with $\mathrm{Norm}(\cdot)$ denoting the Gaussian distribution and $\boldsymbol{I}_K$ being the $K \times K$ dimensional identity matrix, the noise variable $\boldsymbol{\epsilon}_n^{(c,m)} \in \mathbb{R}^L \sim \mathrm{Norm}(0, (\mathrm{diag}(\boldsymbol{\alpha}^{(c,m)}))^{-1})$ in which $\mathrm{diag}(\boldsymbol{\alpha}^{(c,m)})$ is a diagonal variance matrix with the diagonal elements $\boldsymbol{\alpha}^{(c,m)} = \{\alpha_l^{(c,m)}\}_{l=1}^{L} \in \mathbb{R}^+$ being different, then the nth HRRP sample $\boldsymbol{x}^{(c,m)}(n) \in \mathbb{R}^L \sim \mathrm{Norm}(\boldsymbol{\mu}^{(c,m)}, \boldsymbol{A}^{(c,m)}\boldsymbol{A}^{(c,m)\mathrm{T}} + (\mathrm{diag}(\boldsymbol{\alpha}^{(c,m)}))^{-1})$ with $n \in \{1, \ldots, N\}$ and N denoting the number of HRRP samples in this frame.

As discussed in [6], the FA model assumes that the sample $\boldsymbol{x}^{(c,m)}(n)$ is drawn approximately from a low-rank Gaussian, which means that $\boldsymbol{A}^{(c,m)}\boldsymbol{A}^{(c,m)\mathrm{T}} = \sum_{j=1}^{J} \zeta_j^{(c,m)} \boldsymbol{v}_j^{(c,m)} \boldsymbol{v}_j^{(c,m)\mathrm{T}}$, with the orthonormal $\boldsymbol{v}_j \in \mathbb{R}^L$,

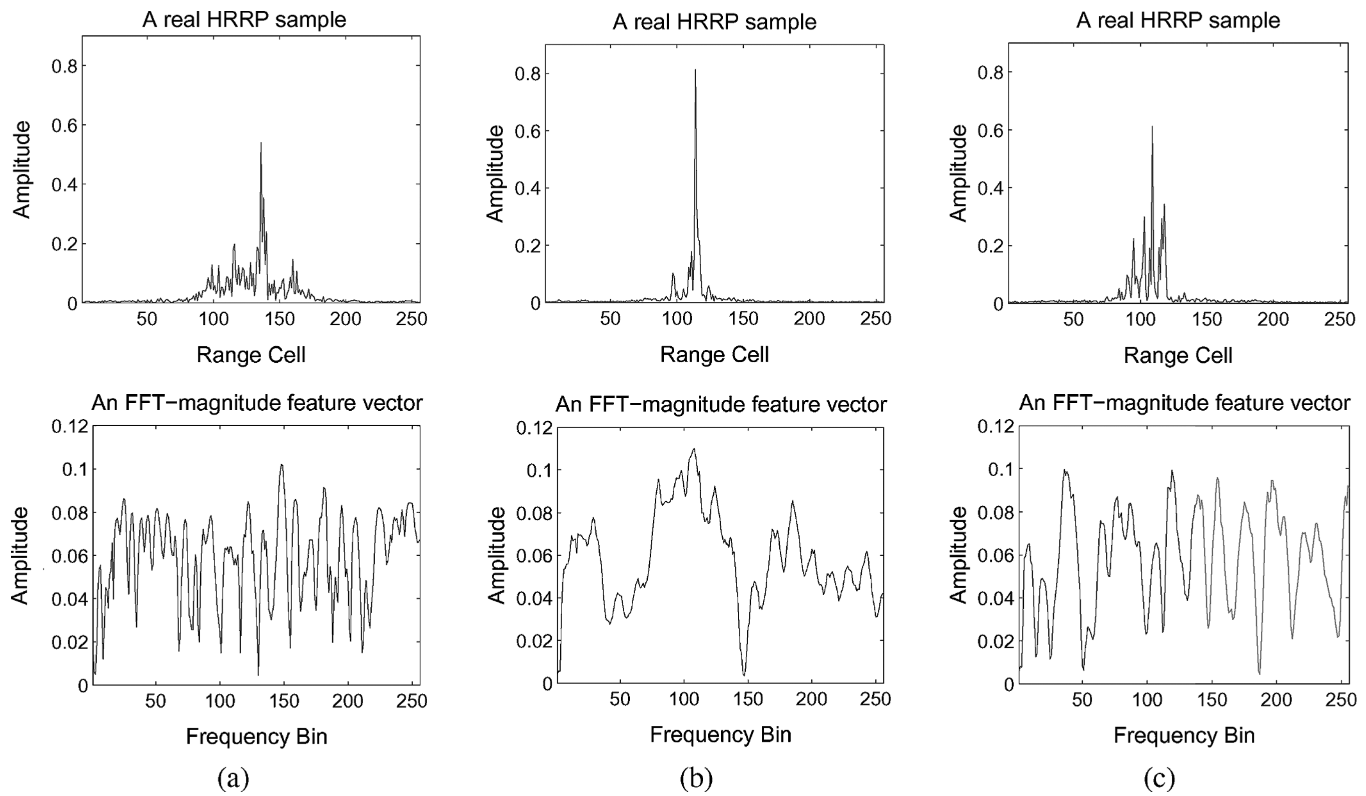

Fig. 2. Real HRRP samples and corresponding FFT-magnitude feature vectors of different airplane targets at the target-aspect angle of 165°. (a) Yak-42. (b) Cessna Citation S/II. (c) An-26.

singular values $\zeta_j \in \mathbb{R}^+$, $\zeta_1 \geqslant \zeta_2 \cdots \geqslant \zeta_J$, $\{1/\alpha_l^{(c,m)}\}_{l=1}^L$ being small relative to ζ_J, and the rank J smaller than the sample dimensionality, i.e., $J < L$. Then the low-rank $\mathrm{Norm}(\boldsymbol{\mu}^{(c,m)}, \boldsymbol{A}^{(c,m)}\boldsymbol{A}^{(c,m)\mathrm{T}} + (\mathrm{diag}(\boldsymbol{\alpha}^{(c,m)}))^{-1})$ with the unknown parameters $\boldsymbol{\Theta}^{(c,m)} = \{\boldsymbol{\mu}^{(c,m)}, \boldsymbol{A}^{(c,m)}, \boldsymbol{\alpha}^{(c,m)}\}$ is the distribution model for the frame dataset $\{\boldsymbol{x}^{(c,m)}(n)\}_{n=1}^N$. In [8], we obtained the maximum-likelihood (ML) estimators of the unknown parameters via an iterative expectation-maximization (EM) algorithm. In the training phase, we learned $\sum_{c=1}^C M_c$ independent FA models based on the whole training dataset from all C targets. If the learning of models for data subsets separately is termed single task learning (STL), the above model is referred to as the STL-FA model.

2) MTL-FA Model Construction: As discussed in Section I, in order to obtain the robust estimators of the unknown parameters in a parametric statistical model, we usually use the training data as much as possible. Therefore, in [8], we used $N = 1024$ HRRP samples from each frame to learn the STL-FA model. However, in the real application, it is hard to collect enough training data for all target-aspects of each target, due to the sampling rate limitation. Rather than building models for each data subset associated with different target-aspect individually, it is desirable to appropriately share the information among these related data. Thus we wish to construct a MTL-FA model for all M_c frames from target c jointly.

To avoid the time-shift compensation between different target-aspect frames, which cannot be dealt with via the conventional time-shift compensation methods due to the scatterers' MTRC between different target-aspect frames, here the MTL-FA model will be developed for the time-shift invariant feature vectors introduced in Section II.A rather than the real HRRP data.

Moreover, in [8], the rank J, i.e., number of factors, is preset and fixed, which may lead to overfitting if set too high or underfitting if set too low. Recent papers [6], [19], and [26] propose to use the truncated Beta process (BP) for inferring J in the FA model with application to compressive sensing, high-dimensional data analysis and image processing, respectively. Based on these works, a similar nonparametric structure is considered here to incorporate the model selection for the number of factors into the MTL-FA structure.

Our hierarchical model for the cth target with $c = 1, 2, \ldots, C$ may be represented as

$$\begin{aligned}
&\boldsymbol{y}^{(c,m)}(n) = \boldsymbol{A}^{(c)}\boldsymbol{w}_n^{(c,m)} + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\epsilon}_n^{(c,m)}\\
&\boldsymbol{w}_n^{(c,m)} = \hat{\boldsymbol{w}}_n^{(c,m)} \circ \boldsymbol{z}^{(c,m)}\\
&\boldsymbol{z}^{(c,m)} = \left[z_1^{(c,m)}, z_2^{(c,m)}, \ldots, z_K^{(c,m)}\right]^{\mathrm{T}}\\
&\hat{\boldsymbol{w}}_n^{(c,m)} \sim \mathrm{Norm}(0, \gamma^{-1}\boldsymbol{I}_K), \quad \gamma \sim \mathrm{Ga}(d_0, e_0)\\
&z_k^{(c,m)} \sim \mathrm{Ber}\left(\pi_k^{(c)}\right)\\
&\pi_k^{(c)} \sim \mathrm{Beta}(a_0/K, b_0(K-1)/K)\\
&\boldsymbol{A}_{\cdot k}^{(c)} \sim \mathrm{Norm}\left(\boldsymbol{0}, \frac{1}{L}\boldsymbol{I}_L\right),\\
&\boldsymbol{\mu}^{(c,m)} \sim \mathrm{Norm}\left(\boldsymbol{\mu}_0^{(c,m)}, \frac{1}{N}\boldsymbol{I}_L\right)\\
&\boldsymbol{\epsilon}_n^{(c,m)} \sim \mathrm{Norm}\left(\boldsymbol{0}, \mathrm{diag}\left(1/\alpha_1^{(c,m)}, 1/\alpha_2^{(c,m)}, \ldots, 1/\alpha_L^{(c,m)}\right)\right)\\
&\alpha_f^{(c,m)} \sim \mathrm{Ga}(g_0, h_0)\\
&\qquad n = 1, 2, \ldots, N; \quad m = 1, 2, \ldots, M_c\\
&\qquad k = 1, 2, \ldots, K; \quad f = 1, 2, \ldots, L
\end{aligned} \tag{5}$$

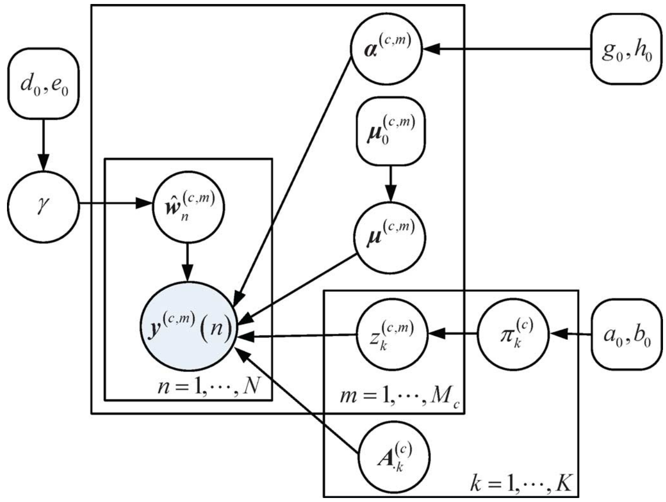

Fig. 3. Graphical representation of the MTL-FA model for the target c with $c = 1, 2, \ldots, C$. The detailed generative process is described in (5). Here k is the factor index, c is the target index, n is the sample index in a frame, m is the frame index, and $\boldsymbol{y}^{(c,m)}(n)$ denotes the FFT-magnitude feature vector of the nth HRRP sample in the mth frame of target c. For the symbol K, N or M_c at the edge of a box indicates that there is a stack of such boxes for the index k, n, or m, where K denotes the number of factors, N denotes the number of samples in a frame, M_c denotes the number of frames for the target c.

where

$$\boldsymbol{y}^{(c,m)}(n) = [|\mathcal{F}_1^{(c,m)}(n)|, |\mathcal{F}_2^{(c,m)}(n)|, \ldots, |\mathcal{F}_L^{(c,m)}(n)|]^{\mathrm{T}}$$

denotes the nth L-length FFT-magnitude feature vector in the mth frame of the cth target, L denotes the number of frequency bins in a FFT-magnitude feature vector, and N denotes the number of samples in the frame; $\boldsymbol{\mu}^{(c,m)} \in \mathbb{R}^L$ is the mean variable of the mth frame of the cth target; $\boldsymbol{A}^{(c)} = [\boldsymbol{A}_{\cdot 1}^{(c)}, \boldsymbol{A}_{\cdot 2}^{(c)}, \ldots, \boldsymbol{A}_{\cdot K}^{(c)}] \in \mathbb{R}^{L \times K}$ is the shared factor loading matrix of the cth target with $\boldsymbol{A}_{\cdot k}^{(c)}$ for $k = 1, \ldots, K$ denoting the kth column of $\boldsymbol{A}^{(c)}$ and K being an integer chosen to be large relative to the number of anticipated factors (e.g., we may set $K = 2 \times L$ in the following experiments); $\boldsymbol{w}_n^{(c,m)} = \hat{\boldsymbol{w}}_n^{(c,m)} \circ \boldsymbol{z}^{(c,m)} \in \mathbb{R}^K$ is the latent factor variable of the mth frame of the cth target, with $\circ$ representing the point-wise (Hadamard) vector product and $\boldsymbol{z}^{(c,m)}$ being a binary vector for selecting the columns of $\boldsymbol{A}^{(c)}$ for the mth frame; $\boldsymbol{\epsilon}_n^{(c,m)} \in \mathbb{R}^L$ is the noise variable of the mth frame of the cth target, with the diagonal variance matrix $\mathrm{diag}(1/\alpha_1^{(c,m)}, 1/\alpha_2^{(c,m)}, \ldots, 1/\alpha_L^{(c,m)})$; a_0, b_0, d_0, e_0, g_0, h_0, and $\{\boldsymbol{\mu}_0^{(c,m)}\}_{m=1}^{M_c}$ are preset hyperparameters. We use $\mathrm{Norm}(\cdot)$, $\mathrm{Ga}(\cdot)$, $\mathrm{Ber}(\cdot)$, and $\mathrm{Beta}(\cdot)$ to refer, respectively, to Gaussian, Gamma, Bernoulli and Beta densities parameterized by the arguments inside parentheses. A graphical representation of this model is depicted in Fig. 3.

Compared to the STL-FA model in [8], the main differences in the model structure come from three aspects as follows.

First, (5) is a hierarchical model with some prior distributions on the parameters or the distributions of the parameters in the original STL-FA in [8]. According to the Bayesian theory [2], more accurate parameter estimation can be obtained via the reasonable prior distribution setting for the case of small data learning.

Second, we use the BP structure for inferring the number of factors in (5). For finite K, the expected number of nonzero components in $\boldsymbol{z}^{(c,m)}$ is $a_0 K / [a_0 + b_0(K-1)]$. When a_0 and b_0 are small relative to K as in the following experiments, most components in $\boldsymbol{z}^{(c,m)}$ will be zero, which means that the binary vector $\boldsymbol{z}^{(c,m)}$ is sparse. According to (5), $\boldsymbol{y}^{(c,m)}(n) \sim \mathrm{Norm}(\boldsymbol{\mu}^{(c,m)}, \boldsymbol{\Sigma}^{(c,m)})$ and

$$\boldsymbol{\Sigma}^{(c,m)} = \boldsymbol{A}^{(c)} \left[\gamma^{-1} \left(\mathrm{diag}\left(\boldsymbol{z}^{(c,m)}\right)\right)\right] \boldsymbol{A}^{(c)\mathrm{T}} + \left(\mathrm{diag}\left(N \cdot \mathbf{1} + \boldsymbol{\alpha}^{(c,m)}\right)\right)^{-1} \quad (6)$$

where $\mathbf{1} = [1, 1, \ldots, 1]^{\mathrm{T}} \in \mathbb{R}^{L \times 1}$. Although there are fixed K candidate columns in $\boldsymbol{A}^{(c)}$, i.e., K candidate factors, only a subset of the candidate columns (factors) are selected via the sparseness-inducing properties of the BP construction, which best fit the data as quantified via the likelihood. When performing Bayesian inference, the posterior density function on $\boldsymbol{z}^{(c,m)}$ defines the number of columns in $\boldsymbol{A}^{(c)}$ that contribute to the FA model, and hence this yields the approximate rank of $\boldsymbol{\Sigma}^{(c,m)}$ in (6), i.e., the number of factors J for the mth frame of target c.

Third, the factor loading matrix is shared across all aspect-frames from a given target, but the latent factor variables are target-aspect dependent in (5). The reason why the factor loading matrix $\boldsymbol{A}^{(c)}$ is selected to be shared is that in (5) each column of $\boldsymbol{A}^{(c)}$, i.e., $\boldsymbol{A}_{\cdot k}^{(c)}$, represents a basis and a sample $\boldsymbol{y}^{(c,m)}(n)$ is the linear combination of the bases selected via $\boldsymbol{z}^{(c,m)}$ with the mixing weights represented by $\hat{\boldsymbol{w}}_n^{(c,m)}$. Thus the model structure implies that the bases selected by different target-aspect frames are learned jointly via the feature data of these frames. Similar sharing structures (*not exactly same*) are applied to compressive sensing and image processing problems in [6] and [26]. By contrast with learning a unique factor loading matrix for each frame in [8], such MTL mechanism offers the potential to more accurately estimate the factor loading matrix via more related data. The experimental results in Section IV-C will further justify our sharing structure.

III. Model Learning and Classification

A. Posterior Inference

In the training phase, we seek to estimate the posterior distribution of the latent variables $\boldsymbol{\Psi}$ in the model, given the observed training data $\boldsymbol{Y}$ and the hyper-parameters $\boldsymbol{\Upsilon}$, via Bayesian inference. In the proposed hierarchical model, the density function at one layer is conjugate to the density function at the layer above it, and therefore inference may be implemented via a variational Bayesian (VB) [1] or Gibbs-sampling analysis [3], with analytic update equations. Here we employ VB inference as a compromise between accuracy and efficiency.

In VB inference, we seek a variational distribution $q(\boldsymbol{\Psi}^{(c)})$ to approximate the true posterior distribution of the latent variables $p(\boldsymbol{\Psi}^{(c)} \,|\, \boldsymbol{Y}^{(c)}, \boldsymbol{\Upsilon}^{(c)})$. The expression

$$\log p\left(\boldsymbol{Y}^{(c)} \middle| \boldsymbol{\Upsilon}^{(c)}\right) = \mathcal{L}\left(q\left(\boldsymbol{\Psi}^{(c)}\right)\right) + \mathcal{KL}\left(q\left(\boldsymbol{\Psi}^{(c)}\right) \middle\| \, p\left(\boldsymbol{\Psi}^{(c)} \middle| \boldsymbol{Y}^{(c)}, \boldsymbol{\Upsilon}^{(c)}\right)\right) \quad (7)$$

where

$$\mathcal{L}(q(\boldsymbol{\Psi}^{(c)})) = \int q(\boldsymbol{\Psi}^{(c)}) \log \frac{p(\boldsymbol{Y}^{(c)}|\boldsymbol{\Psi}^{(c)}, \boldsymbol{\Upsilon}^{(c)})p(\boldsymbol{\Psi}^{(c)}|\boldsymbol{\Upsilon}^{(c)})}{q(\boldsymbol{\Psi}^{(c)})} \mathrm{d}\boldsymbol{\Psi}^{(c)}$$

forms a lower bound. From (7) $\log p(\boldsymbol{Y}^{(c)} \,|\, \boldsymbol{\Upsilon}^{(c)}) \geq \mathcal{L}(q(\boldsymbol{\Psi}^{(c)}))$, since the Kullback-Leibler (KL) divergence $\mathcal{KL}(q(\boldsymbol{\Psi}^{(c)}) \,\|\, p(\boldsymbol{\Psi}^{(c)} \,|\, \boldsymbol{Y}^{(c)}, \boldsymbol{\Upsilon}^{(c)})) = \int q(\boldsymbol{\Psi}^{(c)}) \log \frac{q(\boldsymbol{\Psi}^{(c)})}{p(\boldsymbol{\Psi}^{(c)} \,|\, \boldsymbol{Y}^{(c)}, \boldsymbol{\Upsilon}^{(c)})} \mathrm{d}\boldsymbol{\Psi}^{(c)}$ is nonnegative, which measures the similarity between the two distributions. Accordingly, the goal of minimizing $\mathcal{KL}(q(\boldsymbol{\Psi}^{(c)}) \,\|\, p(\boldsymbol{\Psi}^{(c)} \,|\, \boldsymbol{Y}^{(c)}, \boldsymbol{\Upsilon}^{(c)}))$, i.e., maximizing the similarity between the variational distribution and the true posterior, reduces to adjusting $q(\boldsymbol{\Psi}^{(c)})$ to maximize $\mathcal{L}(q(\boldsymbol{\Psi}^{(c)}))$.

VB inference [1] assumes a factorized $q(\boldsymbol{\Psi}^{(c)})$, i.e., $q(\boldsymbol{\Psi}^{(c)}) = \prod_h q_h(\Psi_h^{(c)})$, where $\Psi_h^{(c)}$ represents each latent variable, and $q_h(\Psi_h^{(c)})$ is the corresponding variational distribution typically with the same form as employed in $p(\boldsymbol{\Psi}^{(c)} \,|\, \boldsymbol{Y}^{(c)}, \boldsymbol{\Upsilon}^{(c)})$. With such an assumption, the variational distributions can be updated iteratively to increase the lower bound $\mathcal{L}(q(\boldsymbol{\Psi}^{(c)}))$. A general method for performing variational inference for conjugate-exponential Bayesian networks is as follows: For a given node in a graphic model, write out the posterior as though everything were known, take the logarithm, the expectation with respect to all unknown parameters, and exponentiate the result. The algorithm iteratively updates $q(\boldsymbol{\Psi}^{(c)})$ so that $\mathcal{L}(q(\boldsymbol{\Psi}^{(c)}))$ approaches to $\log p(\boldsymbol{Y}^{(c)} \,|\, \boldsymbol{\Upsilon}^{(c)})$ until convergence.

The VB-based Bayesian inference for the MTL-FA model (5) in the training stage is summarized in Algorithm 1. The algorithm starts from a set of initial values $\boldsymbol{\Psi}_0^{(c)}$, and when the lower bound variation rate between two iterations is lower than a small positive constant ς, we terminate the algorithm.

Algorithm 1: MTL-FA model learning for the FFT-magnitude feature set of complex HRRP data from all target-aspects of a given target in the training phase via VB inference

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Input: The FFT-magnitude feature set of the cth target $\boldsymbol{Y}^{(c)}$; the hyper-parameters $\boldsymbol{\mu}_0^{(c,m)} = (1/N) \sum_{n=1}^{N} \boldsymbol{y}^{(c,m)}(n)$, $a_0 = b_0 = 1$, $d_0 = e_0 = g_0 = h_0 = 10^{-6}$; and the threshold of the lower bound variation rate ς.

Initialization: $\boldsymbol{\Psi}^{(c)} \leftarrow \boldsymbol{\Psi}_0^{(c)}$; set $\tau = 0$ and $\mathcal{L}(\tau) = -\infty$.

repeat

$\tau = \tau + 1$

% mean variable

$q(\boldsymbol{\mu}^{(c,m)}) = \mathrm{Norm}(\boldsymbol{t}^{(c,m)}, \boldsymbol{\Gamma}^{(c,m)})$, $\boldsymbol{\Gamma}^{(c,m)} = [N + N\mathrm{diag}(\langle \boldsymbol{\alpha}^{(c,m)} \rangle)]^{-1}$

$\boldsymbol{t}^{(c,m)} = \boldsymbol{\Gamma}^{(c,m)}[N\boldsymbol{\mu}_0^{(c,m)} + \mathrm{diag}(\langle \boldsymbol{\alpha}^{(c,m)} \rangle) \sum_{n=1}^{N} (\boldsymbol{y}^{(c,m)}(n) - \langle \boldsymbol{A}^{(c)} \rangle \mathrm{diag}\,(\langle \boldsymbol{z}^{(c,m)} \rangle) \langle \hat{\boldsymbol{w}}_n^{(c,m)} \rangle)]$

% factor loading matrix

$q(A_{fk}^{(c)}) = \mathrm{Norm}(\xi_{fk}^{(c)}, \lambda_{fk}^{(c)})$, $\lambda_{fk}^{(c)} = [L + \sum_{m=1}^{M_c} \langle \alpha_f^{(c,m)} \rangle \langle z_k^{(c,m)} \rangle \sum_{n=1}^{N} \langle \hat{w}_{kn}^{(c,m)2} \rangle]^{-1}$

$\xi_{fk}^{(c)} = \lambda_{fk}^{(c)} (\sum_{m=1}^{M_c} \langle \alpha_f^{(c,m)} \rangle \langle z_k^{(c,m)} \rangle \sum_{n=1}^{N} \langle \hat{w}_{kn}^{(c,m)} \rangle \langle y_f^{(c,m)(-k)}(n) \rangle)$

$y_f^{(c,m)(-k)}(n) = y_f^{(c,m)}(n) - \sum_{h \neq k} A_{fh}^{(c)} z_h^{(c,m)} \hat{w}_{hn}^{(c,m)} - \mu_f^{(c,m)}$

% latent factor variables

$q(z_k^{(c,m)}) = \mathrm{Ber}\left(\frac{\exp(p_{k1}^{(c,m)})}{\exp(p_{k0}^{(c,m)}) + \exp(p_{k1}^{(c,m)})} \right)$

$$\begin{cases} p_{k1}^{(c,m)} = \langle \ln \pi_k^{(c)} \rangle - [\frac{1}{2} \sum_{n=1}^{N} (\langle \hat{w}_{kn}^{(c,m)2} \rangle \langle \boldsymbol{A}_{\cdot k}^{(c)\mathrm{T}} \mathrm{diag}(\boldsymbol{\alpha}^{(c,m)}) \boldsymbol{A}_{\cdot k}^{(c)} \rangle - \tilde{p}_{k1}^{(c,m)}(n))] \\ p_{k0}^{(c,m)} = \langle \ln(1 - \pi_k^{(c)}) \rangle \end{cases}$$

$\tilde{p}_{k1}^{(c,m)}(n) = 2\langle \boldsymbol{y}^{(c,m)(-k)}(n) \rangle^{\mathrm{T}} \mathrm{diag}(\langle \boldsymbol{\alpha}^{(c,m)} \rangle) \langle \boldsymbol{A}_{\cdot k}^{(c)} \rangle \langle \hat{w}_{kn}^{(c,m)} \rangle$

$q(\hat{\boldsymbol{w}}_n^{(c,m)}) = \mathrm{Norm}(\boldsymbol{\rho}_n^{(c,m)}, \boldsymbol{B}_n^{(c,m)})$, $\boldsymbol{B}_n^{(c,m)} = [\boldsymbol{D}^{(c,m)} + \langle \gamma \rangle \boldsymbol{I}_k]^{-1}$

$\boldsymbol{\rho}_n^{(c,m)} = \boldsymbol{B}_n^{(c,m)} [(\langle \boldsymbol{A}^{(c)} \rangle^{\mathrm{T}} \circ \langle \boldsymbol{Z}^{(c,m)} \rangle) \mathrm{diag}(\langle \boldsymbol{\alpha}^{(c,m)} \rangle)(\boldsymbol{y}^{(c,m)}(n) - \langle \boldsymbol{\mu}^{(c,m)} \rangle)]$

$\boldsymbol{D}^{(c,m)} = \langle (\boldsymbol{A}^{(c)\mathrm{T}} \circ \boldsymbol{Z}^{(c,m)}) \mathrm{diag}(\boldsymbol{\alpha}^{(c,m)})(\boldsymbol{A}^{(c)} \circ \boldsymbol{Z}^{(c,m)\mathrm{T}}) \rangle$, $\boldsymbol{Z}^{(c,m)} = [\boldsymbol{z}^{(c,m)}, \ldots, \boldsymbol{z}^{(c,m)}] \in \mathbb{R}^{K \times L}$

$q(\pi_k^{(c)}) = \mathrm{Beta}(a_0/K + \sum_{m=1}^{M} \langle z_k^{(c,m)} \rangle, b_0(K - 1)/K + M - \sum_{m=1}^{M} \langle z_k^{(c,m)} \rangle)$

$q(\gamma) = \mathrm{Ga}(d_0 + \frac{KM_cN}{2}, e_0 + \frac{1}{2} \sum_{m=1}^{M_c} \sum_{n=1}^{N} \langle \hat{\boldsymbol{w}}_n^{(c,m)\mathrm{T}} \hat{\boldsymbol{w}}_n^{(c,m)} \rangle)$

% noise precision variable

$q(\alpha_f^{(c,m)}) = \mathrm{Ga}(g_0 + N/2, h_0 + (1/2) \sum_{n=1}^{N} \langle [y_f^{(c,m)}(n) - \boldsymbol{A}_{f\cdot}^{(c)} (\boldsymbol{z}^{(c,m)} \circ \hat{\boldsymbol{w}}_n^{(c,m)}) - \mu_f^{(c,m)}]^2 \rangle)$

% lower bound

$\mathcal{L}(\tau) = \int q(\boldsymbol{\Psi}^{(c)}) \log \frac{p(\boldsymbol{Y}^{(c)} \,|\, \boldsymbol{\Psi}^{(c)}, \boldsymbol{\Upsilon}^{(c)}) p(\boldsymbol{\Psi}^{(c)} \,|\, \boldsymbol{\Upsilon}^{(c)})}{q(\boldsymbol{\Psi}^{(c)})} \mathrm{d}\boldsymbol{\Psi}^{(c)}$

until $\mathcal{L}(\tau) - \mathcal{L}(\tau - 1) < \varsigma \mathcal{L}(\tau - 1)$.

In Algorithm 1, $\langle \cdot \rangle$ means the posterior expectation for the latent variable over the corresponding distribution on it; $\boldsymbol{A}_{f\cdot}^{(c)}$ denotes the fth row of $\boldsymbol{A}^{(c)}$; $A_{fk}^{(c)}$ is the kth component of $\boldsymbol{A}_{f\cdot}^{(c)}$; $\hat{w}_{kn}^{(c,m)}$ represents the kth component of $\hat{\boldsymbol{w}}_n^{(c,m)}$, $y_f^{(c,m)}(n)$ represents the fth component of $\boldsymbol{y}^{(c,m)}(n)$, and $\mu_f^{(c,m)}$ represents the fth component of $\boldsymbol{\mu}^{(c,m)}$; $\langle \ln \pi_k^{(c)} \rangle = \psi(a_0/K + \sum_{m=1}^{M} \langle z_k^{(c,m)} \rangle) - \psi([a_0 + b_0(K-1)]/K + M)$ and $\langle \ln(1 - \pi_k^{(c)}) \rangle = \psi(b_0(K-1)/K + M - \sum_{m=1}^{M} \langle z_k^{(c,m)} \rangle) - \psi([a_0 + b_0(K - 1)]/K + M)$ with $\psi(\cdot)$ denoting the Digamma function; for $\boldsymbol{D}^{(c,m)}$, we have to consider it as two parts: the off-diagonal elements of $\boldsymbol{D}^{(c,m)}$ are $(\langle \mathbf{A}^{(c)} \rangle^{\mathrm{T}} \circ \langle \boldsymbol{Z}^{(c,m)} \rangle) \mathrm{diag}(\langle \boldsymbol{\alpha}^{(c,m)} \rangle)(\langle \boldsymbol{A}^{(c)} \rangle \circ \langle \mathbf{Z}^{(c,m)} \rangle^{\mathrm{T}})$, and the diagonal elements are $[\sum_{f=1}^{L} \langle A_{fk}^{(c)2} \rangle \langle \alpha_f^{(c,m)} \rangle] \langle z_k^{(c,m)} \rangle$, where $\boldsymbol{Z}^{(c,m)} = [\boldsymbol{z}^{(c,m)}, \ldots, \boldsymbol{z}^{(c,m)}] \in \mathbb{R}^{K \times L}$ denotes a matrix of the column vector $\boldsymbol{z}^{(c,m)}$ repeated L times; $\boldsymbol{\mu}_0^{(c,m)} = (1/N)\sum_{n=1}^{N} \boldsymbol{y}^{(c,m)}(n)$ is the maximum-likelihood (ML) estimator of the mean variable $\boldsymbol{\mu}^{(c,m)}$, which can lead to fast convergence for $\boldsymbol{\mu}^{(c,m)}$; the noninformative hyper-parameters are employed for other variables, by setting $a_0 = b_0 = 1$ and $d_0 = e_0 = g_0 = h_0 = 10^{-6}$, same as the corresponding priors used in [6] and [19].

B. Noise-Robust Classification

Our model differs from the dictionary learning model (referred to as BPFA model) for image processing [26] in that the binary vector $\boldsymbol{z}^{(c,m)} = \{z_k^{(c,m)}\}_{k=1}^{K}$ in (5) is target-aspect dependent, not sample dependent, which means $\boldsymbol{z}^{(c,m)}$ is shared across all the samples from the corresponding frame $\{\boldsymbol{y}^{(c,m)}(n)\}_{n=1}^{N}$. In [26], the BPFA model is used as a feature extraction method and the sample dependent factor variable is the extracted feature vector. For classification, a supervised classifier, such as support vector machine (SVM) or relevance vector machine (RVM), need be trained via the extracted factor variables from the training data, and the class membership of a test sample is determined based on the classifier output with the new factor variable extracted from the test sample as the input measurement. As discussed in Section I, Bayes classifier is used in the statistical recognition scheme, with which the class membership of a test sample is determined based on the maximum class-conditional likelihood searched from those of all classes. Considering the target-aspect sensitivity of HRRP data, here we develop a statistical model for each target-aspect frame, and the class-conditional likelihood of each class will be obtained via substituting the test sample into all of the target-aspect dependent statistical models and searching the maximum one. Therefore, by contrast with the classification scheme used in [26], the statistical recognition scheme used here can avoid estimating or extracting the factor variable for a new test sample in the classification stage.

After learning the model for each target c with $c = 1, 2, \ldots, C$, $q(\boldsymbol{\Psi}^{(c)})$ is obtained. If a test HRRP sample $\boldsymbol{x}^*$ is measured under the same SNR circumstance as that of training data, according to (5), the frame-conditional likelihood of the corresponding FFT-magnitude feature $\boldsymbol{y}^*$ given the frame m of the target c is

$$p(\boldsymbol{y}^* \,|\, m, c) = \mathrm{Norm}\left(\boldsymbol{\chi}^{(c,m)}, \boldsymbol{\Omega}^{(c,m)}\right) \tag{8}$$

with

$$\begin{cases} \boldsymbol{\chi}^{(c,m)} = \mathbb{E}\left[\boldsymbol{A}^{(c)}\left(\hat{\boldsymbol{w}}_*^{(c,m)} \circ \boldsymbol{z}^{(c,m)}\right) + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\epsilon}_*^{(c,m)}\right] \\ \qquad\;\; = \mathbb{E}\left[\boldsymbol{A}^{(c)}\left(\hat{\boldsymbol{w}}_*^{(c,m)} \circ \boldsymbol{z}^{(c,m)}\right) + \boldsymbol{\mu}^{(c,m)}\right] \\ \boldsymbol{\Omega}^{(c,m)} = \mathbb{C}\left[\boldsymbol{A}^{(c)}\left(\hat{\boldsymbol{w}}_*^{(c,m)} \circ \boldsymbol{z}^{(c,m)}\right) + \boldsymbol{\mu}^{(c,m)} + \boldsymbol{\epsilon}_*^{(c,m)}\right] \end{cases} \tag{9}$$

where $\mathbb{E}[\cdot]$ and $\mathbb{C}[\cdot]$ represent the expectation and covariance operators, respectively. If we use the prior distribution for $\hat{\boldsymbol{w}}_*^{(c,m)}$, then $\boldsymbol{\chi}^{(c,m)} = \langle \boldsymbol{\mu}^{(c,m)} \rangle$ and $\boldsymbol{\Omega}^{(c,m)} = \mathbb{C}[\boldsymbol{A}^{(c)}(\hat{\boldsymbol{w}}_*^{(c,m)} \circ \boldsymbol{z}^{(c,m)})] + \mathbb{C}[\boldsymbol{\mu}^{(c,m)}] + \mathbb{C}[\boldsymbol{\epsilon}_*^{(c,m)}] = \langle \boldsymbol{A}^{(c)}[\frac{e_0}{d_0}\mathrm{diag}(\boldsymbol{z}^{(c,m)})]\boldsymbol{A}^{(c)\mathrm{T}} \rangle + \langle (\boldsymbol{\mu}^{(c,m)} - \langle \boldsymbol{\mu}^{(c,m)} \rangle)(\boldsymbol{\mu}^{(c,m)} - \langle \boldsymbol{\mu}^{(c,m)} \rangle)^{\mathrm{T}} \rangle + \mathrm{diag}(\langle \boldsymbol{\alpha}^{(c,m)} \rangle)^{-1}$, where the prior mean for $\hat{\boldsymbol{w}}_*^{(c,m)}$ is 0 and the prior covariance for $\hat{\boldsymbol{w}}_*^{(c,m)}$ is e_0/d_0. However, this estimator may not be accurate since $\hat{\boldsymbol{w}}_*^{(c,m)}$ has its own posterior distribution. Thus, we use the posterior distributions of the corresponding frame data $\{q(\hat{\boldsymbol{w}}_n^{(c,m)})\}_{n=1}^{N}$ to estimate the posterior mean and covariance for $\hat{\boldsymbol{w}}_*^{(c,m)}$ as follows:

$$\begin{cases} \mathbb{E}\left[\hat{\boldsymbol{w}}_*^{(c,m)}\right] = \frac{1}{N}\sum_{n=1}^{N}\left\langle \hat{\boldsymbol{w}}_n^{(c,m)} \right\rangle \\ \mathbb{C}\left[\hat{\boldsymbol{w}}_*^{(c,m)}\right] = \frac{1}{N}\sum_{n=1}^{N}\left\langle \overline{\hat{\boldsymbol{w}}^{(c,m)}}_n \overline{\hat{\boldsymbol{w}}^{(c,m)}}_n^{\mathrm{T}} \right\rangle \end{cases} \tag{10}$$

where $\overline{\hat{\boldsymbol{w}}^{(c,m)}}_n = \hat{\boldsymbol{w}}_n^{(c,m)} - \langle \hat{\boldsymbol{w}}_n^{(c,m)} \rangle$. Paper [6] used the similar estimator from Gibbs sampling for the compressive sensing application.

Since the frame models from a given target are assumed to be independent [10], then the class-conditional likelihood of target c ($c \in \{1, 2, \ldots, C\}$) is $p(\boldsymbol{y}^* \,|\, c) = \max_m p(\boldsymbol{y}^* \,|\, m, c)$ with $m \in \{1, 2, \ldots, M_c\}$. According to Bayes theorem $p(c \,|\, \boldsymbol{y}^*) = \frac{p(\boldsymbol{y}^* \,|\, c)p(c)}{\sum_{c=1}^{C} p(\boldsymbol{y}^* \,|\, c)p(c)}$ with $c = 1, \ldots, C$, the class membership of $\boldsymbol{y}^*$ can be assigned to the class with the maximum posterior probability, i.e., $c^* = \arg\max_c p(c \,|\, \boldsymbol{y}^*)$. Similar frame matching method is also utilized in [8], [10], and [12].

In the above discussion, we assume the noise level of the test sample is same as those of the training samples, the statistical recognition method is referred to as *MTL-FA model without modification in classification* in this paper. However, for RATR problem, the training data can usually be obtained under high SNR via some cooperative measurement experiments or directly via simulations; while the test samples are usually got in the noncooperative circumstance (e.g., at the battle time), where the high SNR cannot be guaranteed due to the large distance between the noncooperative targets and radar.

If we assume that the measured test sample under low SNR can be expressed as $\tilde{\boldsymbol{y}}^* = \boldsymbol{y}^* + \boldsymbol{\nu}^*$, where the noise level of $\boldsymbol{y}^*$ is same as those of the training samples (namely with high SNR) and $\boldsymbol{\nu}^*$ is the new noise variable introduced from the test condition with the assumption that $\boldsymbol{\nu}^* \sim \mathrm{Norm}(\boldsymbol{0}, r_*^{-1}\boldsymbol{I}_L)$, we can have

$$p(\tilde{\boldsymbol{y}}^* \,|\, m, c) = \mathrm{Norm}\left(\boldsymbol{\chi}^{(c,m)}, \tilde{\boldsymbol{\Omega}}^{(c,m)}\right) \tag{11}$$

with

$$\tilde{\boldsymbol{\Omega}}^{(c,m)} = \boldsymbol{\Omega}^{(c,m)} + r_*^{-1}\boldsymbol{I}_L \tag{12}$$

Therefore, if we can estimate the test noise level r_* via some preprocessing methods, the matched target indicator c^* and frame indicator m^* can be easily obtained via the above frame matching method based on $\{p(\tilde{\boldsymbol{y}}^* \,|\, m, c)\}_{m=1,c=1}^{M_c,C}$. The noise considered here mainly refers to the heat noise from the receiver, which can be assumed to be stationary, accordingly, there are several ways to accurately estimate the noise power (i.e., noise variance r_*^{-1}) [21], of which a simple approach is to measure the noise power from the receiver with no input signal. Since this noise-robust classification method utilizes the test noise level information r_*, the statistical recognition method is referred to as *noise-robust MTL-FA model with the test noise level prior* in this paper.

Another approach of the noise-robust classification based on MTL-FA model is to adaptively learn a new noise variable with the different noise precision variable given the noised test sample $\tilde{\boldsymbol{y}}^*$ for each frame of each target. Given the outputs, $\{q(\boldsymbol{\Psi}^{(c)})\}_{c=1}^{C}$, $\{\boldsymbol{\alpha}_*^{(c,m)}\}_{c=1,m=1}^{C,M_c}$ can be inferred for $\tilde{\boldsymbol{y}}^*$, using a VB inference algorithm that iterates between $\hat{\boldsymbol{w}}_*^{(c,m)}$ and $\boldsymbol{\alpha}_*^{(c,m)}$ with $c = 1, 2, \ldots, C$ and $m = 1, 2, \ldots, M_c$. The equations are similar to those in Algorithm 1, with the inference for other variables removed. This statistical recognition method is referred to as *noise-robust MTL-FA model with the adaptively learned test noise level* in this paper, and the corresponding classification stage is summarized in Algorithm 2, where c^* and m^* represent the matched target indicator and frame indicator.

Algorithm 2: Classification phase of the noise-robust MTL-FA model with the adaptively learned test noise level for the FFT-magnitude feature of complex HRRP data via VB inference

Input: The FFT-magnitude feature vector of a noised HRRP test sample $\tilde{\boldsymbol{y}}^*$, the learned posterior distribution of the latent variables in the MTL-FA model of each target $q(\boldsymbol{\Psi}^{(c)})$ ($c \in \{1, 2, \ldots, C\}$), the hyper-parameters $g_0 = h_0 = 10^{-6}$, and the threshold of the lower bound variation rate ς.

For $c = 1$ **to** C

Initialization: $\boldsymbol{\alpha}_*^{(c,m)} \leftarrow \boldsymbol{\alpha}_0^{(c,m)}$; set $\tau = 0$ and $\mathcal{L}(\tau) = -\infty$

repeat

$\tau = \tau + 1$

% latent factor component

$q(\hat{\boldsymbol{w}}_*^{(c,m)}) = \mathrm{Norm}(\boldsymbol{\rho}_*^{(c,m)}, \boldsymbol{B}_*^{(c,m)})$, $\boldsymbol{B}_*^{(c,m)} = [\boldsymbol{D}^{(c,m)} + \langle\gamma\rangle\boldsymbol{I}_k]^{-1}$

$\boldsymbol{\rho}_*^{(c,m)} = \boldsymbol{B}_*^{(c,m)}[(\langle\boldsymbol{A}^{(c)}\rangle^{\mathrm{T}} \circ \langle\boldsymbol{Z}^{(c,m)}\rangle)\mathrm{diag}(\langle\boldsymbol{\alpha}_*^{(c,m)}\rangle) (\tilde{\boldsymbol{y}}^* - \langle\boldsymbol{\mu}^{(c,m)}\rangle)]$

$\boldsymbol{D}^{(c,m)} = \langle(\boldsymbol{A}^{(c)\mathrm{T}} \circ \boldsymbol{Z}^{(c,m)})\mathrm{diag}(\boldsymbol{\alpha}_*^{(c,m)})(\boldsymbol{A}^{(c)} \circ \boldsymbol{Z}^{(c,m)\mathrm{T}})\rangle$, $\boldsymbol{Z}^{(c,m)} = [\boldsymbol{z}^{(c,m)}, \ldots, \boldsymbol{z}^{(c,m)}] \in \mathbb{R}^{K \times L}$

% noise precision variable

$q(\alpha_{*f}^{(c,m)}) = \mathrm{Ga}(g_0 + 1/2, h_0 + (1/2) \langle[\tilde{y}_f^* - \boldsymbol{A}_{f\cdot}^{(c)}(\boldsymbol{z}^{(c,m)} \circ \hat{\boldsymbol{w}}_*^{(c,m)}) - \mu_f^{(c,m)}]^2\rangle)$

% lower bound

$\mathcal{L}(\tau) = \int q(\boldsymbol{\Psi}_*^{(c)}) \log \frac{p(\tilde{\boldsymbol{y}}^* \mid \boldsymbol{\Psi}_*^{(c)}, \boldsymbol{\Upsilon}^{(c)}) p(\boldsymbol{\Psi}_*^{(c)} \mid \boldsymbol{\Upsilon}^{(c)})}{q(\boldsymbol{\Psi}_*^{(c)})} \mathrm{d}\boldsymbol{\Psi}_*^{(c)}$

until $\mathcal{L}(\tau) - \mathcal{L}(\tau - 1) < \varsigma\mathcal{L}(\tau - 1)$.

For $m = 1$ **to** M_c

$p(\tilde{\boldsymbol{y}}^* \mid m, c) = \mathrm{Norm}(\boldsymbol{\chi}^{(c,m)}, \tilde{\boldsymbol{\Omega}}^{(c,m)})$

$\boldsymbol{\chi}^{(c,m)} = \langle\boldsymbol{A}^{(c)}\rangle\mathrm{diag}(\langle\boldsymbol{z}^{(c,m)}\rangle)\mathbb{E}[\hat{\boldsymbol{w}}_*^{(c,m)}] + \langle\boldsymbol{\mu}^{c,m}\rangle$

$\tilde{\boldsymbol{\Omega}}^{(c,m)} = \langle\boldsymbol{A}^{(c)}\mathrm{diag}(\boldsymbol{z}^{(c,m)})\mathbb{C}[\hat{\boldsymbol{w}}_*^{(c,m)}]\mathrm{diag}(\boldsymbol{z}^{(c,m)})\boldsymbol{A}^{(c)\mathrm{T}}\rangle + \mathbb{C}[\boldsymbol{\mu}^{(c,m)}] + \mathrm{diag}(\langle\boldsymbol{\alpha}_*^{(c,m)}\rangle)^{-1}$

End

$p(\tilde{\boldsymbol{y}}^*|c) = \max_m p(\tilde{\boldsymbol{y}}^*|m, c)$, $m^* = \arg\max_m p(\tilde{\boldsymbol{y}}^*|m, c)$

End

$c^* = \arg\max_c p(c \mid \tilde{\boldsymbol{y}}^*) = \arg\max_c \frac{p(\tilde{\boldsymbol{y}}^* \mid c)p(c)}{\sum_{c=1}^{C} p(\tilde{\boldsymbol{y}}^* \mid c)p(c)}$.

Furthermore, the condition distribution for $\boldsymbol{y}^*$ given $\tilde{\boldsymbol{y}}^*$, c^* and m^* may be evaluated analytically as

$$\begin{aligned} p(\boldsymbol{y}^* \mid \tilde{\boldsymbol{y}}^*, c^*, m^*) &= \frac{p(\boldsymbol{y}^*)p(\tilde{\boldsymbol{y}}^* \mid \boldsymbol{y}^*)}{\int p(\boldsymbol{y}^*)p(\tilde{\boldsymbol{y}}^* \mid \boldsymbol{y}^*)\mathrm{d}\boldsymbol{y}^*} \\ &= \mathrm{Norm}\left(\hat{\boldsymbol{\chi}}^{(c^*,m^*)}, \hat{\boldsymbol{\Omega}}^{(c^*,m^*)}\right) \end{aligned} \tag{13}$$

with

$$\begin{cases} \hat{\boldsymbol{\Omega}}^{(c^*,m^*)} = \boldsymbol{\Omega}^{(c^*,m^*)} - \boldsymbol{\Omega}^{(c^*,m^*)}\tilde{\boldsymbol{\Omega}}^{(c^*,m^*)-1}\boldsymbol{\Omega}^{(c^*,m^*)} \\ \hat{\boldsymbol{\chi}}^{(c^*,m^*)} = \boldsymbol{\Omega}^{(c^*,m^*)}\tilde{\boldsymbol{\Omega}}^{(c^*,m^*)-1}\left(\tilde{\boldsymbol{y}}^* - \boldsymbol{\chi}^{(c^*,m^*)}\right) + \boldsymbol{\chi}^{(c^*,m^*)}. \end{cases} \tag{14}$$

Therefore, we can get the posterior expectation (estimation) of the feature vector with high SNR as $\langle\boldsymbol{y}^*\rangle = \hat{\boldsymbol{\chi}}^{(c^*,m^*)}$, which can be further utilized to other applications.

Here we should state that the difference between the three classification methods discussed in this section, i.e., the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the test noise level prior and the noise-robust MTL-FA model with the adaptively learned test noise level, is how to use the MTL-FA model trained under high SNR to discriminate the test data collected under low SNR in the classification stage, and the training stage is completely same for the three methods, i.e., Algorithm 1.

IV. Experimental Results

A. Measured Data

We examine the performance of the proposed model on the measured HRRP data from three real airplanes, which are also used in [8]–[12], [20]. The parameters of the airplane targets and measurement radar are shown in Table I, and the projections of target trajectories onto the ground plane are shown in Fig. 4, from which the aspect angle of an airplane can be estimated according to its relative position to radar. As shown in Fig. 4, the measured data are segmented, where each segment contains 25600 HRRP samples and each sample is a 256-length vector, i.e., $L = 256$.

In order to test the generalization performance of the recognition methods, there are two requirements for choosing the training data and test data. *a)* The training data cover almost all of the target-aspect angles of the test data; *b)* The elevation angles of the test data are different from those of the training data. Therefore, data from the 2nd segment and a part of the 5th segment of Yak-42, the 6th and the 7th segments of Cessna Citation S/II, the 5th and the 6th segments of An-26 are taken as

TABLE I
PARAMETERS OF PLANES AND RADAR IN THE ISAR EXPERIMENT

radar parameters	center frequency		5520 MHz
	bandwidth		400 MHz
planes	length (m)	width (m)	height (m)
Yak-42	36.38	34.88	9.83
Cessna Citation S/II	14.40	15.90	4.57
An-26	23.80	29.20	9.83

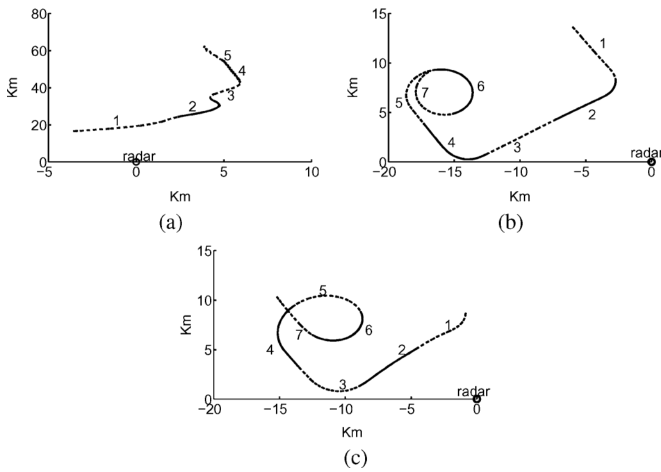

Fig. 4. Projections of target trajectories onto the ground plane. (a) Yak-42. (b) Cessna Citation S/II. (c) An-26.

the training samples, and other data segments are taken as test samples in our experiments.

It is a prerequisite for RATR based on complex HRRP data to deal with the initial-phase, time-shift, target-aspect, and amplitude-scale sensitivity problems [11]. As discussed in Section II-A, the proposed FFT-magnitude feature is invariant to the initial-phase and time-shift. We have discussed the target-aspect sensitivity issue in Section II-B. Same as those in our previous work [8]–[11], [20], we use the aspect-frame partition method to deal with the target-aspect sensitivity of the training dataset. There are 35 aspect-frames for Yak-42, 50 for Cessna Citation S/II, and 50 for An-26, of which each aspect-frame totally has 1024 HRRP samples, i.e., $C = 3$, $M_1 = 35$, $M_2 = M_3 = 50$, $N = 1024$. In the following experiments, we will compare the recognition performance via training the proposed statistical model with different training data sizes, i.e., different numbers of training samples from each frame (different N). In addition, since the intensity of an HRRP is a function of radar transmitting power, target distance, radar antenna gain, radar receiver gain, radar system losses and so on, HRRPs measured by different radars or under different conditions will have different amplitude-scales. To deal with this amplitude-scale sensitivity, each FFT-magnitude feature vector extracted from the original complex HRRP sample is normalized by dividing its L_2 norm, $\bar{\boldsymbol{y}}(n) = \frac{\boldsymbol{y}(n)}{\|\boldsymbol{y}(n)\|_2} = \frac{\boldsymbol{y}(n)}{\sqrt{\boldsymbol{y}^{\mathrm{T}}(n)\boldsymbol{y}(n)}}$.

In the following experiments via our MTL-FA model, in the training phase, the FFT-magnitude feature vectors extracted from complex HRRPs in each aspect-frame are amplitude L_2 normalized; while in the classification phase, the single normalized feature vector is tested one by one and the whole test dataset is the same as those used in [8], [10], and [12]. The recognition rate in the following experimental results is obtained by dividing the total number of test samples into the number of correctly classified test samples.

This HRRP dataset is measured under high SNR. As discussed in Sections I and III-B, for RATR problem, we usually assume that the training data are under high SNR and the test samples are got under the different condition of low SNR. For aircraft-like targets under a look-up scenario such as the case of this dataset, the noises in the inphase and quadrature echoes of targets can be assumed to be Gaussian white noises. Therefore, in order to evaluate the robustness of our proposed method, we add simulated white noises to the inphase and quadrature components of the raw complex HRRP test data. The SNR is defined as

$$\begin{aligned}\mathrm{SNR} &= 10 \times \log_{10}\left(\frac{\bar{P}_{\boldsymbol{x}}}{P_{\mathrm{Noise}}}\right)\\ &= 10 \times \log_{10}\left(\frac{\sum_{l=1}^{L} P_{x_l}}{L \times P_{\mathrm{Noise}}}\right) \quad (15)\end{aligned}$$

where $\bar{P}_{\boldsymbol{x}}$ denotes the average power of original HRRP, P_{x_l} denotes the power of the original echo from the lth range cell, L denotes the number of range cells (here $L = 256$), and P_{Noise} denotes the power of noise. Thus $10^{\frac{\mathrm{SNR}}{10}}$ equals to the rate of the average power of original HRRP to the power of the noise.

The hyper-parameter values in the MTL-FA model are set as in Algorithms 1 and 2. In the following experiments, we set the truncation level for Beta process prior at $K = 2 \times L = 2 \times 256 = 512$ (namely the number of columns in $\boldsymbol{A}^{(c)}$), and the threshold of the lower bound variation rate to be $\varsigma = 10^{-5}$.

B. Recognition Performance

The recognition accuracies of the STL-FA model [8], the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the test noise level prior, and the noise-robust MTL-FA model with the adaptively learned test noise level, versus the size of training dataset for the test data under the condition of $\mathrm{SNR} = 15$ dB are shown in Fig. 5. All the three MTL-FA methods are trained via Algorithm 1 with the training data under high SNR, but the classification methods are different for the test data under the condition of $\mathrm{SNR} = 15$ dB as discussed in Section III-B. Here the STL-FA model is directly used for the real HRRP samples and the three MTL-FA models are for the FFT-magnitude feature extracted from the complex HRRP data. There are five training datasets: 64×135, 128×135, 256×135, 512×135, 1024×135, where 64×135 means that 64 feature vectors randomly selected from each of the 135 aspect-frames, i.e., totally 8640 training samples, similarly for other data size expressions. The recognition rates in Fig. 5 are averaged over ten runs, and the error bars denote the corresponding standard deviations.

From Fig. 5, we can clearly see that the usage of the accurate test noise level prior or the adaptively learned test noise level in the classification stage can improve the recognition performance of the MTL-FA model trained under high SNR. In addition, for our MTL-FA model, when the training data size

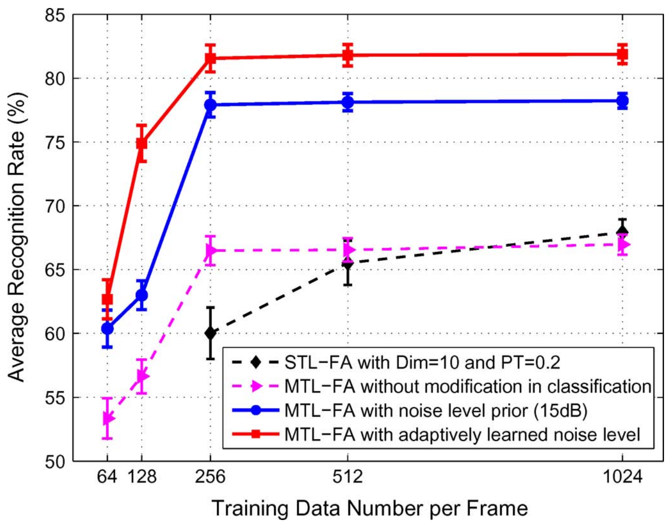

Fig. 5. Variation of the recognition performance with training data size, via the STL-FA model with the dimensionality of the latent factor variable (i.e., number of factors) $J = 10$ and power transformation $p = 0.2$ [8], the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the test noise level prior, and the noise-robust MTL-FA model with the adaptively learned test noise level, for the test data under the condition of $\mathrm{SNR} = 15$ dB. All the three MTL-FA methods are trained via Algorithm 1 with the training data under high SNR, but the classification methods are different for the test data under the condition of $\mathrm{SNR} = 15$ dB as discussed in Section III-B. Here the STL-FA model is directly used for the real HRRP samples and the three MTL-FA models are for the FFT-magnitude feature extracted from the complex HRRP data. The recognition rates are averaged over ten runs, and the error bars denote the corresponding standard deviations.

is larger than 256×135, the recognition performance generally remains stable; while for the STL-FA model the training data size of each target-aspect frame is required to be larger than the length of the feature vector due to the full covariance estimation in the separate frame model learning, i.e., $N \geq L$ with here $L = 256$, and the average recognition rate of the STL-FA model decreases about 7% when the training data size decreases from 1024×135 to 256×135 as shown in Fig. 5. Thus the better learning performance with small data size is an advantage of MTL over STL. The reason why the recognition rates obtained via the modification with the accurate test noise level prior are lower than those via the adaptive modification under the test condition of $\mathrm{SNR} = 15$ dB will be explained in the following paragraph after showing the variation of the recognition performance with SNR in Fig. 6.

Fig. 6 depicts the average recognition rates versus SNR, generated via the five statistical recognition methods, the dynamic MTL-TSB-HMM model [12], the STL-FA model [8], the MTL-FA model without modification in classification and two versions of the noise-robust MTL-FA model, i.e., the noise-robust MTL-FA model with the test noise level prior, and the noise-robust MTL-FA model with the adaptively learned test noise level. All these methods are trained with the training data under high SNR, but the classifications are implemented under different SNR conditions. Here the first two models are directly used for the real HRRP samples and the three MTL-FA models are for the FFT-magnitude feature extracted from the complex HRRP data. For brevity in Fig. 6, we omit the recognition curves from the three traditional statistical models for the real HRRP data, i.e., Gaussian model [14], [18], Gamma model

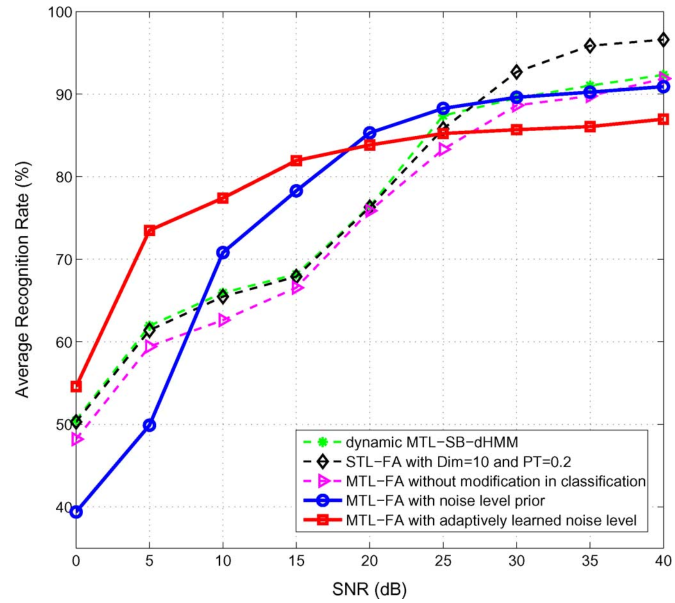

Fig. 6. Variation of the recognition performance with SNR, via the dynamic MTL-TSB-HMM [12], the STL-FA model with the dimensionality of the latent factor variable (i.e., number of factors) $J = 10$ and power transformation $p = 0.2$ [8], the MTL-FA model without modification in classification, the noise-robust MTL-FA model with the test noise level prior, and the noise-robust MTL-FA model with the adaptively learned test noise level. All these methods are trained with the training data under high SNR, but the classifications are implemented under different SNR conditions. Here the first two models are directly used for the real HRRP samples and the three versions of our MTL-FA model are for the FFT-magnitude feature extracted from the complex HRRP data. Moreover, our MTL-FA model and STL-FA model are trained with $1024 \times 135 = 138240$ training data; while the dynamic MTL model with $256 \times 135 = 34560$ training data. The recognition rates are averaged over ten runs.

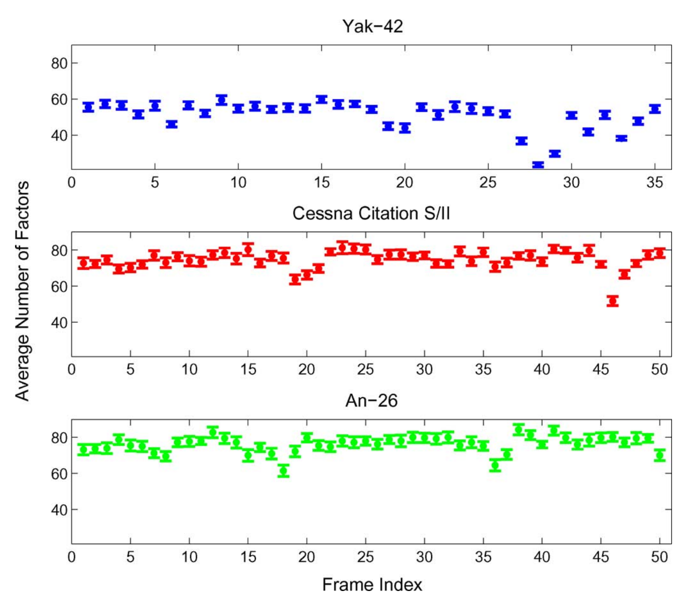

Fig. 7. Number of factors per aspect-frame for 256 training samples per frame, estimated via our MTL-FA model. The numbers of factors are averaged over 10 runs of the model, and the error bars denote the corresponding standard deviations.

[7] and the two-distribution compounded model [10]. From [12], the STL-FA model and the dynamic MTL-TSB-HMM model obviously outperform the three traditional models, especially under the condition of low SNR. In this recognition experiment, our MTL-FA model and STL-FA model are trained

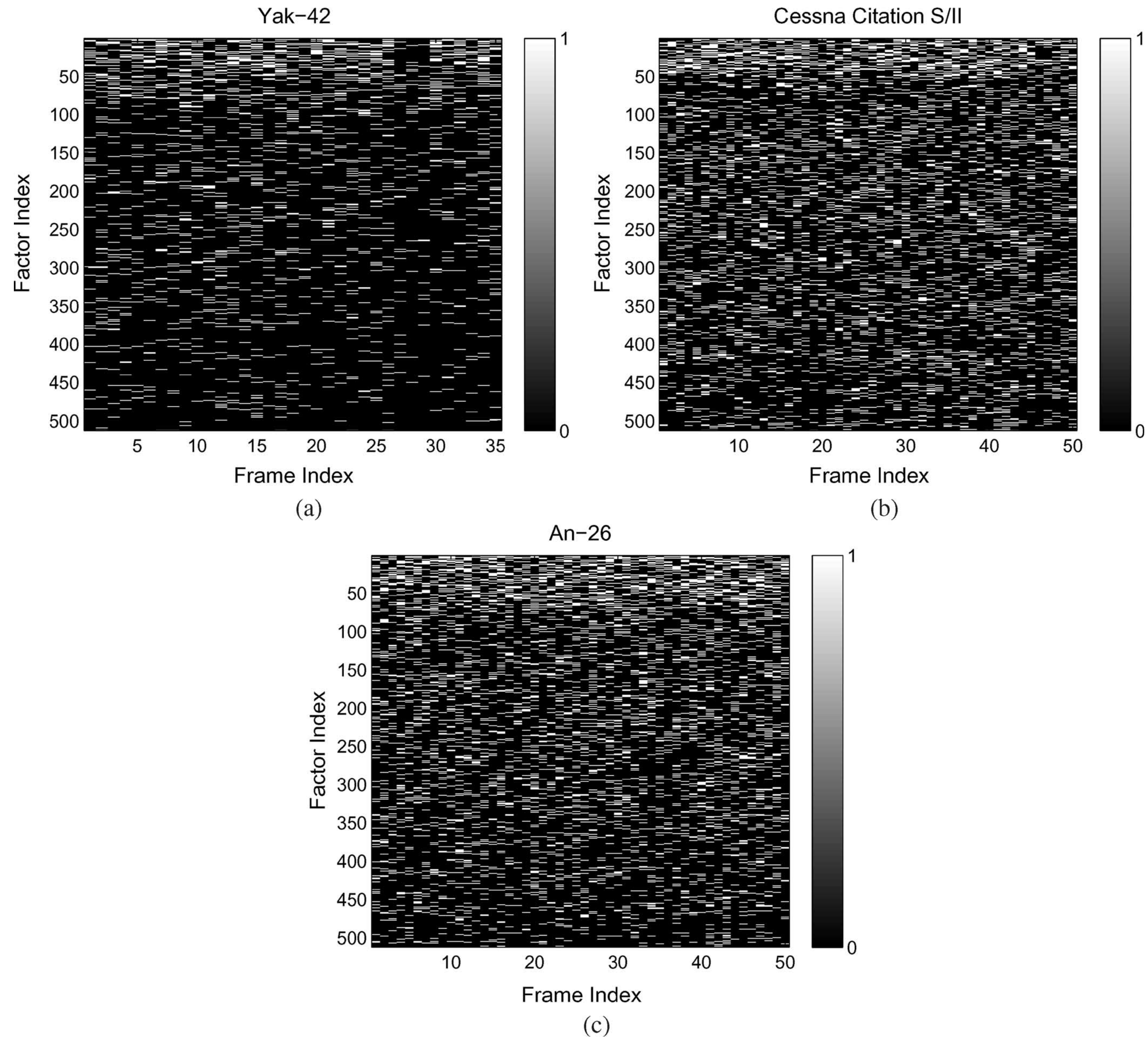

Fig. 8. Learned factor occupancy, i.e., $\boldsymbol{Z}^{(c)} = [\langle \boldsymbol{z}^{(c,1)} \rangle, \langle \boldsymbol{z}^{(c,2)} \rangle, \ldots, \langle \boldsymbol{z}^{(c,M_c)} \rangle]$ with $c = 1, 2, 3$, $M_1 = 35$ and $M_2 = M_3 = 50$, for 256 training samples per frame, generated via a run of our MTL-FA model. (a) Yak-42: $\boldsymbol{Z}^{(1)}$. (b) Cessna Citation S/II: $\boldsymbol{Z}^{(2)}$. (c) An-26: $\boldsymbol{Z}^{(3)}$.

with $1024 \times 135 = 138240$ training data; while the dynamic MTL model with $256 \times 135 = 34560$ training data. Here it is important to point out that our MTL-FA model with smaller training data size, e.g., 512×135 or 256×135, can obtain the similar recognition performance as shown in Fig. 6, which has been demonstrated via the experiment shown in Fig. 5; while the performance of the STL-FA model will dramatically worsen when the training data size decreases also as demonstrated via the experiment shown in Fig. 5.

As shown in Fig. 6, when $\mathrm{SNR} \geqslant 30$ dB, the STL-FA model with power transformation outperforms other four models; when $\mathrm{SNR} \geqslant 20$ dB, our noise-robust MTL-FA model with the test noise level prior is superior to our noise-robust MTL-FA model with the adaptively learned test noise level; when SNR is 25 or 20 dB, the noise-robust MTL-FA model with the test noise level prior is better than other methods; when $\mathrm{SNR} \leqslant 15$ dB, the noise-robust MTL-FA model with the adaptively learned test noise level yields the best recognition rates. The reason why the accurate noise level prior cannot yield the best performance when $\mathrm{SNR} \leqslant 15$ dB, which can also be observed in Fig. 5 for $\mathrm{SNR} = 15$ dB, is that the noise components in HRRP data are complex Gaussian distributed as discussed in Section IV-A, but the FA model assumes the noise a real Gaussian variable. Although we still use the FA structure in the adaptive method, the relearned posterior distribution of the noise precision variable can adjust the statistical model to match the noised test data better than that directly modified by adding a real noise component as in the method using the test noise level prior. If we assume that the average recognition rate threshold for the real application is 70%, our noise-robust MTL-FA model with the adaptively learned test noise level can work under the test condition of $\mathrm{SNR} \geqslant 5$ dB; while the

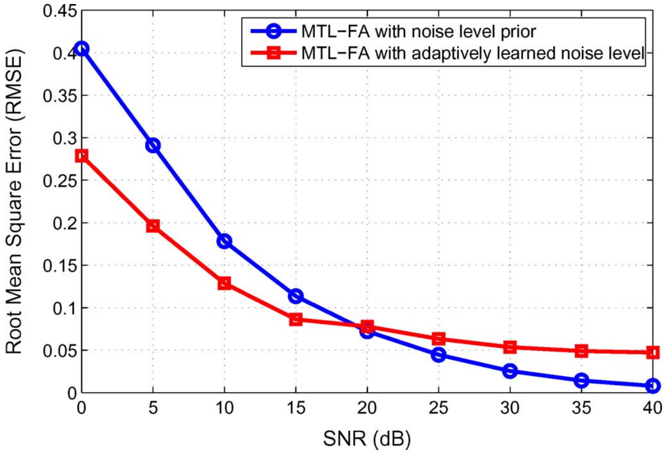

Fig. 9. Variation of the reconstruction errors of the test data with SNR, via our noise-robust MTL-FA model with the test noise level prior, and our noise-robust MTL-FA model with the adaptively learned test noise level, for $256 \times 135 = 34560$ training data. Both the two methods are trained via Algorithm 1 with the training data under high SNR, but the reconstruction methods are different for the noised test data as discussed in Section III-B. Here the average relative reconstruction errors are averaged over ten runs.

STL-FA model and the dynamic MTL-TSB-HMM model need $\mathrm{SNR} \geqslant 20$ dB. Since the SNR of the target echo directly relates to the distance between the target and radar for a given noise power, our noise-robust methods can significantly increase the recognition distance between the target and radar in the real application.

C. Results Analysis

As discussed in Section II-B, the proposed nonparametric model can infer the proper number of factors J for each aspect-frame of the training data. Fig. 7 presents the model selection results for J, where $J = \sum_{k=1}^{K} \langle z_k^{(c,m)} \rangle$ for $m = 1, 2, \ldots, M_c$ and $c = 1, 2, 3$, with $\langle z_k^{(c,m)} \rangle$ denoting the posterior expectation of the binary variable $z_k^{(c,m)}$. The numbers of factors are averaged over ten runs of our MTL-FA model with 256 training data per frame, and the error bars denote the corresponding standard deviations over the ten runs. Although we initialized the truncation level to $K = 512$, there are no more than 90 factors with the corresponding $\langle z_k^{(c,m)} \rangle$ being nonzero for any frame from the three targets, which justifies using the truncated BP prior for our data. In addition, the inferred J varies across the frame index, which shows the utility of using such nonparametric model selection method. By contrast, we fix the same J for all frames in the STL-FA model, which may lead to overfitting or underfitting for some frames. Furthermore, Fig. 8 shows $\langle z_k^{(c,m)} \rangle$ in the same experiment. For each target c, if factor k is used for the frame m, then the posterior probability on this factor usage, $\langle z_k^{(c,m)} \rangle$, should be one; otherwise, it will be zero. Each column in each subfigure of Fig. 8 denotes the inferred factor usages for each frame of the corresponding target. From Fig. 8, we can find the different frames of a given target may select a same factor, which demonstrates it is reasonable to share $\boldsymbol{A}^{(c)}$ across all aspect-frames from a given target.

As discussed in Algorithm 2, the proposed noise-robust methods can also reconstruct the feature vector with high SNR from the noised one. The reconstruction performance quantified via the average relative reconstruction errors is shown in Fig. 9 and examples of reconstruction for an FFT-magnitude feature vector from Yak-42 HRRP data are given in Fig. 10. In Fig. 9, the average relative reconstruction error is defined as $\frac{1}{N^*} \sum_{n=1}^{N^*} \frac{\|\langle \boldsymbol{y}_n^* \rangle - \boldsymbol{y}_n^*\|_2}{\|\boldsymbol{y}_n^*\|_2}$, where $\boldsymbol{y}_n^*$ denotes the nth original feature vector under high SNR, $\langle \boldsymbol{y}_n^* \rangle$ represents the nth reconstructed feature vector from the noised one, and $N^* = 5200$ denotes the number of test samples. Similar to the recognition performance shown in Fig. 6, our noise-robust MTL-FA model with the adaptively learned test noise level outperforms that with the accurate test noise level prior under the test condition of $\mathrm{SNR} \leqslant 15$ dB in the reconstruction performance. From Fig. 10, we can clearly see the reconstruction performance of our adaptive method is encouraged when the test condition of $\mathrm{SNR} \geqslant 0$ dB, and the reconstructed feature vectors can be further utilized to other applications, e.g., analysis on target characteristic, information fusion, and dataset expansion.

D. Computation Burden

All experiments have been performed in nonoptimized software written in Matlab, on a Pentium PC with 1.50 GHz CPU and 2G RAM. For the measured HRRP data, in the training phase, our MTL-FA model with truncation level $K = 512$ (i.e., Algorithm 1), required averagely about 5 hr for each target with about 60 VB iterations to get convergence. In the classification phase, our noise-robust MTL-FA model with the test noise level prior averagely required 0.5592 second to match a test sample with all frame models, which is similar to that of the STL-FA model. However, since our noise-robust MTL-FA model with the adaptively learned test noise level (i.e., Algorithm 2) needs to relearn the test noise level in the classification phase, its computation cost of classification is much larger. Averagely, the adaptive method required 18.0096 s in the classification phase with about 10 VB iterations to get convergence. Thus the larger time cost in the classification phase is a disadvantage of the adaptive method. The time cost of Algorithm 2 can be reduced with a larger threshold ς, which leads to fewer iterations, but the estimation accuracy of the test noise level and thus the recognition performance might be degraded at the same time. Therefore, in the real application, we need to strike an appropriate balance between time cost and recognition performance. Each computation time given above is averaged over 50 runs for the different training data sizes.

V. Conclusion

We have developed a noise-robust factor analysis model based on multitask learning (noise-robust MTL-FA) for the FFT-magnitude feature extracted from complex HRRP data. The initial-phase and time-shift invariant feature allows the factor loading matrix shared for all target-aspects of each target via the MTL construction (here the multiple tasks are linked to specific aspect-frames of HRRPs from each target). Moreover, the proposed framework can update the noise level parameter in the FA model in the classification phase to adaptively match that of the received test sample and further reconstruct the test sample with high SNR from the originally noised one. The RATR experiments based on measured HRRP data show that the proposed methods

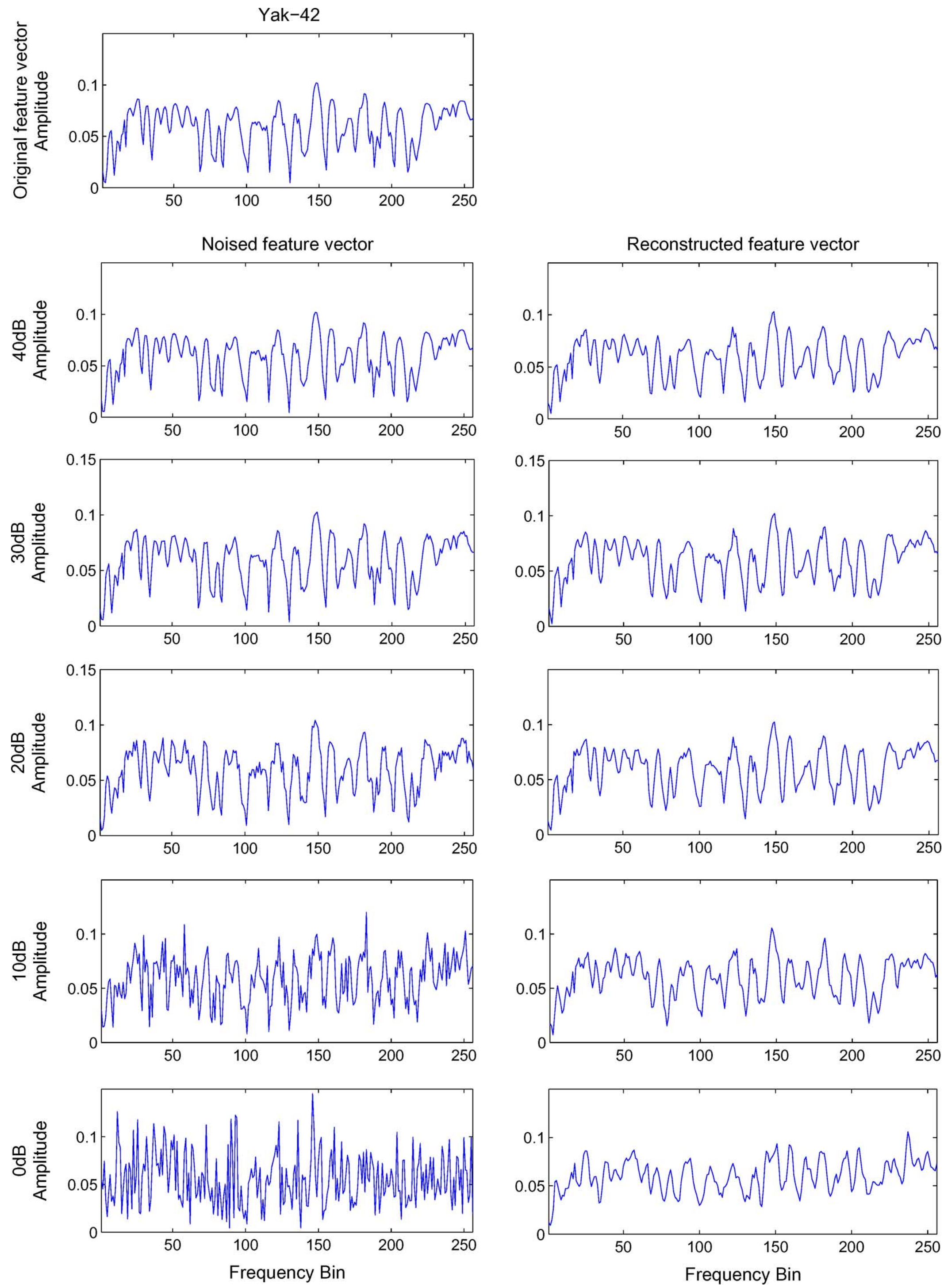

Fig. 10. Reconstruction examples for a FFT-magnitude feature vector from Yak-42 HRRP data, via our noise-robust MTL-FA model with the adaptively learned test noise level. The first row shows the original feature vector, and the subsequent rows show the noised and reconstructed feature vectors under different SNR circumstances (40, 30, 20, 10, and 0 dB).

have inspiring recognition and reconstruction performance with small training data size and under the test condition of low SNR.

REFERENCES

[1] M. J. Beal, "Variational Algorithms for Approximate Bayesian Inference," Ph.D. Dissertation, Gatsby Computational Neuroscience Unit, University College London, , 2003.

[2] C. M. Bishop, *Pattern Recognition and Machine Learning*. New York: Springer Science, 2006.

[3] W. G. Carrara, R. S. Goodman, and R. M. Majewski, *Markov Chain Monte Carlo in Practice*. London, U.K.: Chapman and Hall, 1996.

[4] R. Caruana, "Multitask learning," *Mach. Learn.*, vol. 28, pp. 41–75, 1997.

[5] B. Chen, H. W. Liu, J. Chai, and Z. Bao, "Large margin feature weighting method via linear programming," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 10, pp. 1475–1488, 2009.

[6] M. Chen, J. Silva, J. Paisley, C. Wang, D. Dunson, and L. Carin, "Compressive sensing on manifolds using a nonparametric mixture of factor analyzers: Algorithm and performance bounds," *IEEE Trans. Signal Process.*, vol. 58, no. 12, pp. 6140–6155, 2010.

[7] K. Copsey and A. R. Webb, "Bayesian gamma mixture model approach to radar target recognition," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 39, no. 4, pp. 1201–1217, 2003.

[8] L. Du, H. W. Liu, and Z. Bao, "Radar HRRP statistical recognition: Parametric model and model selection," *IEEE Trans. Signal Process.*, vol. 56, no. 5, pp. 1931–1944, 2008.
[9] L. Du, H. W. Liu, Z. Bao, and M. D. Xing, "Radar HRRP target recognition based on higher-order spectra," *IEEE Trans. Signal Process.*, vol. 53, no. 7, pp. 2359–2368, 2005.
[10] L. Du, H. W. Liu, Z. Bao, and J. Y. Zhang, "A two-distribution compounded statistical model for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 54, no. 6, pp. 2226–2238, 2006.
[11] L. Du, H. W. Liu, Z. Bao, and J. Y. Zhang, "Radar automatic target recognition using complex high-resolution range profiles," in *Proc. IET Radar Sonar Navig.*, 2007, vol. 1, no. 1, pp. 18–26.
[12] L. Du, P. H. Wang, H. W. Liu, M. Pan, F. Chen, and Z. Bao, "Bayesian spatiotemporal multitask learning for radar HRRP target recognition," *IEEE Trans. Signal Process.*, vol. 59, no. 7, pp. 3182–3196, 2011.
[13] W. R. Gilks and D. J. Spiegelhalter, *Spotlight Synthetic Aperture Radar—Signal Processing Algorithms*. Norwood, MA: Arthech House, 1995.
[14] S. P. Jacobs, "Automatic target recognition using high-resolution radar range profiles," Ph.D. dissertation, Washington Univ., St. Louis, MO, 1999.
[15] L. Jouny, F. D. Garber, and R. L. Moses, "Radar target identification using the bispectrum: A comparative study," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 31, no. 1, 1995.
[16] X. J. Liao and Z. Bao, "Circularly integrated bispectra: Novel shift invariant features for high-resolution radar recognition," *Electron. Lett.*, vol. 34, no. 19, pp. 1879–1880, 1998.
[17] X. J. Liao, P. Runkle, and L. Carin, "Identification of ground targets from sequential high-range-resolution radar signatures," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 38, no. 4, pp. 1230–1242, 2002.
[18] R. A. Mitchell and J. J. Westerkamp, "Robust statistical feature based aircraft identification," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 35, no. 3, pp. 1077–1094, 1999.
[19] J. Paisley and L. Carin, "Non-parametric factor analysis with beta process," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2009.
[20] L. Shi, P. H. Wang, H. W. Liu, L. Xu, and Z. Bao, "Radar HRRP statistical recognition with local factor analysis by automatic Bayesian Ying-Yang harmony learning," *IEEE Trans. Signal Process.*, vol. 59, no. 2, pp. 610–617, 2011.
[21] M. I. Skolnik, *Radar Handbook*, 3rd ed. New York: McGraw-Hill, 2008.
[22] A. V. Totsky, I. V. Kurbatov, V. V. Lukin, K. O. Egiazarian, and J. T. Astola, "Combined bispectrum-filtering techniques for radar output signal reconstruction in ATR applications," in *Proc. SPIE*, 2003, vol. 5094, no. 301.
[23] P. H. Wang, F. Z. Dai, M. Pan, L. Du, and H. W. Liu, "Radar HRRP target recognition in frequecny domain based on autoregressive model," in *Proc. IEEE Int. Radar Conf.*, 2011, pp. 714–717.
[24] S. K. Wong, "Non-cooperative target recognition in the frequency domain," in *Proc. IEE Proc.—Radar Sonar Navig.*, 2004, vol. 151, no. 2, pp. 77–84.
[25] X. D. Zhang, Y. Shi, and Z. Bao, "A new feature vector using selected bispectra for signal classification with application in radar target recognition," *IEEE Trans. Signal Process.*, vol. 49, no. 9, pp. 1875–1885, 2001.
[26] M. Zhou, H. Chen, J. Paisley, L. Ren, G. Sapiro, and L. Carin, "Nonparametric Bayesian dictionary learning for sparse image representations," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2009.
[27] F. Zhu, X. D. Zhang, Y. F. Hu, and D. Xie, "Nonstationary hidden Markov models for multiaspect discriminative feature extraction from radar targets," *IEEE Trans. Signal Process.*, vol. 55, no. 5, pp. 2203–2213, 2007.

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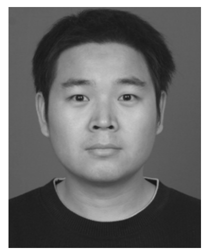

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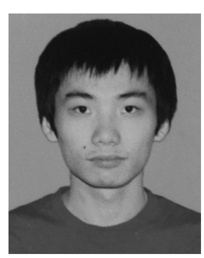

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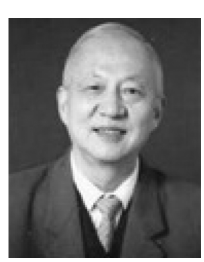

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